

UNIGRAM MODEL

8. For a vocabulary $V = \{v_k\}_K$, consider a unigram model $p = [p_k]_{k=0}^{M-1}$ where n_k is the number of observations of v_k , and $p_k = \frac{n_k}{\sum_K n_k} = \frac{n_k}{N}$

Prove that p is optimal.

→ Proof:

For outcome k , a model assigns probabilities p_k . Each outcome is observed n_k times.

We have to maximize the probability of a set of observed words $w_t \in V$

$$\prod_K p_k^{n_k} \text{ where } n_k = \sum_t w_t = N_k$$

We have to choose p_k - the probability assigned to each word. These probabilities should sum to 1, i.e., $\sum_K p_k = 1$

This is a constrained optimization problem and its solution relies on the method of Lagrange multipliers.

Consider the functions
 $f(p) = \prod_k p_k^{n_k}$ and

$$g(p) = \sum_k p_k - 1 = 0$$

At the point $\hat{p}$ where $g(p) = 0$ and $f(p)$ is maximized, we know that the gradient of f along the surface $g(p) = 0$ is zero (as it is a constant), and the gradient of f along the same surface is zero.

$\therefore \nabla f(p)$ is parallel to $\nabla g(p)$
i.e., there is some scalar λ such that
 $\nabla f(p) = \lambda \nabla g(p)$

Maximizing the log probability $\sum_k n_k \log p_k$

$$\frac{\partial}{\partial p_k} \left(\sum_k n_k \log p_k \right) = \lambda \frac{\partial}{\partial p_k} \left(\sum_k p_k - 1 \right)$$

$$n_k \left(\frac{\partial}{\partial p_k} \sum_k \log p_k \right) = \lambda \frac{\partial}{\partial p_k} \sum_k p_k$$

$$\frac{n_k}{p_k} = \lambda$$

$$\lambda p_k = n_k \quad \text{--- ①}$$

③

Summing on both sides and using the constraint

$$\sum_k p_k = 1$$

$$\sum_k n_k = \lambda \sum_k p_k$$

$$\sum_k n_k = \lambda$$

Plugging this value of λ in ①, we get

$$p_k \sum_k n_k = n_k$$

$$p_k = \frac{n_k}{\sum_k n_k}$$

which is the fraction of all observed words that are v_k .